

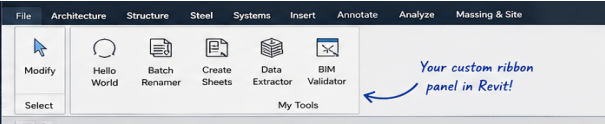
# Revit MCP Server Plugin Development Guide

## Complete Step-by-Step AI-Powered Revit Automation Development Handbook

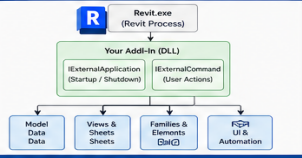


### INTRODUCTION TO REVIT PLUG-IN DEVELOPMENT WITH C# AND THE REVIT API

Build powerful add-ins that automate, extend, and customize Revit.



Your custom ribbon panel in Revit!

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WHAT THE REVIT API GIVES YOU</b> <ul style="list-style-type: none"><li><b>Model Data</b><br/>Access and modify all model elements</li><li><b>Geometry</b><br/>Create, read and manipulate geometry</li><li><b>Parameters</b><br/>Read, write and manage parameters</li><li><b>Views &amp; Sheets</b><br/>Create views, sheets, legends, schedules</li><li><b>Families</b><br/>Load, edit and manage families programmatically</li><li><b>UI Customization</b><br/>Add ribbon panels, buttons, tools and dialogs</li><li><b>Automation</b><br/>Build workflows that save time and reduce manual work</li></ul>                      | <b>DEVELOPMENT ENVIRONMENT</b> <ul style="list-style-type: none"><li><b>Visual Studio</b><br/>(2019 / 2022 / 2022+)</li><li><b>References</b><ul style="list-style-type: none"><li>RevitAPI.dll</li><li>RevitAPIUI.dll</li><li>RevitAPIUIUI.dll (optional)</li></ul></li><li><b>Located in:</b><br/>C:\Program Files\Autodesk\Revit 20XX\</li></ul>                                                                                                                                                                                        | <b>IMPLEMENTING CORE INTERFACES</b> <div><b>IExternalApplication (runs on Revit startup)</b><pre>public class MyApp : IExternalApplication {     public Result OnStartup(UIControlledApplication app)     {         // Create ribbon panel and add buttons         var panel = app.CreateRibbonPanel("My Tools");         var button = panel.AddItem(new PushButtonData(             "HelloWorld", "Hello World",             Assembly.GetExecutingAssembly().Location,             typeof(HelloWorldCommand).FullName));         return Result.Succeeded;     }      public Result OnShutdown(UIControlledApplication app)     {         return Result.Succeeded;     } }</pre></div> <div><b>IExternalCommand (runs when user clicks)</b><pre>public class HelloWorldCommand : IExternalCommand {     public Result Execute(ExternalCommandData commandData,         ref string message, ElementSet elements)     {         TaskDialog.Show("Hello World",             "Hello from your Revit Plug-in! ㊗️");         return Result.Succeeded;     } }</pre></div> |
| <b>BASIC PLUGIN WORKFLOW</b> <ol style="list-style-type: none"><li>Create C# Class Library Project</li><li>Add References to RevitAPI.dll and RevitAPIUI.dll</li><li>Implement IExternalApplication and IExternalCommand</li><li>Create .addin Manifest File</li><li>Place .addin File in Revit Addins Folder (%ProgramData%\Autodesk\Revit\Addins\20XX)</li><li>Launch Revit Plugin loads automatically!</li></ol>                                                                                                                                                                                                                   | <b>EXAMPLE .ADDIN MANIFEST</b> <pre>&lt;?xml version="1.0" encoding="utf-8"?&gt; &lt;RevitAddins&gt;   &lt;Addin Type="Application"&gt;     &lt;Name&gt;My Revit Tools&lt;/Name&gt;     &lt;Assembly&gt;C:\MyAddins\MyRevitTools.dll&lt;/Assembly&gt;     &lt;AddinLocation&gt;20XX-09-09-000-000-000-000\Addins\     &lt;FullClassName&gt;MyNamespace.MyApp&lt;/FullClassName&gt;     &lt;VendorId&gt;MYCO&lt;/VendorId&gt;     &lt;VendorDescription&gt;My Company&lt;/VendorDescription&gt;   &lt;/Addin&gt; &lt;/RevitAddins&gt;</pre> | <b>PLUGIN ARCHITECTURE INSIDE REVIT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>WHAT CAN PLUG-INS AUTOMATE?</b> <ul style="list-style-type: none"><li>Batch operations (rename, delete, update)</li><li>Parameter editing and validation</li><li>Geometry creation and modification</li><li>View, sheet and documentation generation</li><li>BIM data extraction and reporting</li><li>Standards checking and QA/QC</li><li>Custom UI tools and workflows</li><li>Office standards automation</li></ul> <div><p>In short:<br/>If you can do it in Revit manually,<br/>your plugin can probably<br/>do it 100x faster.</p></div> | <b>IMPORTANT NOTES</b> <ul style="list-style-type: none"><li>Plugins run inside Revit, not as standalone web APIs.</li><li>Full Revit installation is required (Revit LT has major API limitations).</li><li>C# is the most stable and recommended language for serious Revit development.</li><li>Test across Revit versions and handle API changes.</li></ul>                                                                                                                                                                            | <b>HELPFUL RESOURCES</b> <ul style="list-style-type: none"><li>Revit API Documentation <a href="https://www.revitapidocs.com">https://www.revitapidocs.com</a></li><li>Autodesk Help - Walkthrough: Hello World <a href="https://help.autodesk.com/view/RVT/2025/ENU/">https://help.autodesk.com/view/RVT/2025/ENU/</a></li><li>Autodesk University - Revit API Classes <a href="https://www.autodesk.com/autodesk-university">https://www.autodesk.com/autodesk-university</a></li><li>Revit API Developer Guide (Official) <a href="https://help.autodesk.com/view/RVT/2025/ENU/">https://help.autodesk.com/view/RVT/2025/ENU/</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <div> Revit API + Your Ideas = Unlimited Possibilities</div> <div> Automate more.</div> <div> Reduce errors.</div> <div> Save hours.</div> <div> Deliver better BIM.</div>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

This professional guide explains how to build a complete Revit MCP Server ecosystem using:

- Revit API
- C# Add-ins
- Python MCP Servers
- AI integrations
- BIM automation workflows
- Enterprise-ready plugin architecture

## 1. System Architecture

The complete system consists of:

- AI Agent / LLM
- MCP Server (Python)
- Revit Add-In (C#)
- Revit API Layer
- BIM Model Interaction

## 2. Required Software

- Autodesk Revit 2024/2025/2026
  - Visual Studio 2022
  - .NET Framework 4.8
  - Python 3.11+
  - VS Code
  - Git
- 

### 3. Project Folder Structure

Recommended structure:

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### 4. Create Revit Add-In

Step-by-step:

1. Create Class Library (.NET Framework)
  2. Add RevitAPI.dll references
  3. Create IExternalApplication
  4. Create Ribbon UI
  5. Create Commands
  6. Build DLL
  7. Create .addin manifest
- 

### 5. Revit API References

Required DLLs:

- C:\Program Files\Autodesk\Revit 2025\  
• RevitAPI.dll  
• RevitAPIUI.dll
- 

### 6. Ribbon UI Workflow

Revit Startup → Plugin Load → Ribbon Creation → Button Commands → API Execution

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### 7. MCP Server Development

Install:

```
pip install fastapi uvicorn pydantic
```

Create a FastAPI server that receives AI commands and routes them to Revit.

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### 8. Communication Methods

Communication between Revit and MCP:

- File-based IPC
  - Named Pipes
  - Local HTTP Server
- 

## 9. Service Architecture

Recommended services:

- GeometryService
  - SheetService
  - ParameterService
  - ExportService
  - ValidationService
- 

## 10. Automation Features

- Automatic wall creation
  - Parameter editing
  - Sheet generation
  - AI-powered BIM commands
  - Documentation automation
  - QA/QC validation
- 

## 11. AI Workflow

User Prompt → AI Interpretation → JSON Command → MCP Server → Revit Add-In → BIM Model

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## 12. Error Handling & Logging

Always use:

- try/catch blocks
  - logging systems
  - transaction management
  - performance tracking
- 

## 13. Enterprise Architecture

AI Layer → MCP Server → API Gateway → Revit Plugin → Revit API → BIM Model

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## 14. Best Practices

- Use modular commands
- Separate UI from logic
- Use services architecture
- Maintain version compatibility
- Keep transactions clean

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## 15. Common Mistakes

- Using .NET Core instead of .NET Framework
- Forgetting transactions
- Hardcoding paths
- No exception handling

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## 16. Learning Roadmap

LEVEL 1: • C# Basics • Revit API Basics • Transactions  
LEVEL 2: • Ribbon UI • Parameters • Geometry API  
LEVEL 3: • Events • Dockable Panels  
LEVEL 4: • MCP + AI Integration • Enterprise Systems

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## 17. Final Goal

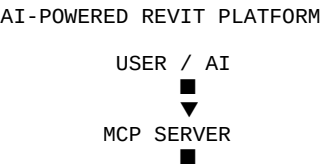
Build:

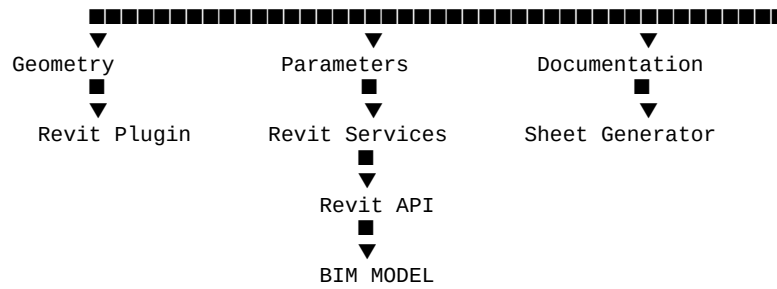
- AI-powered BIM tools
- Intelligent Revit assistants
- Automated documentation systems
- Enterprise BIM automation platforms

## Recommended Folder Structure

```
RevitMCPProject/  
■  
■■■■ RevitAddin/  
■ ■■■■ Commands/  
■ ■■■■ Services/  
■ ■■■■ Models/  
■ ■■■■ UI/  
■ ■■■■ App.cs  
■  
■■■■ MCPServer/  
■ ■■■■ main.py  
■ ■■■■ tools/  
■ ■■■■ routes/  
■ ■■■■ services/  
■  
■■■■ Shared/  
■■■■ Docs/
```

## Final Brain Map





The future of BIM is programmable.

By combining Revit API + MCP + AI systems, you can create intelligent BIM assistants capable of automating modeling, documentation, validation, and enterprise workflows.